Linear Stability To Check

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Step 0: The model

$$\partial_t n = n\,(c_{\text{ext}}(1-l)\,r_{\text{ext}} + c_\times r_\times - m) + D_n \nabla^2 n$$

$$\partial_t r_{\text{ext}} = K - \kappa\, r_{\text{ext}} - c_{\text{ext}}\, r_{\text{ext}}\, n + D_{\text{ext}} \nabla^2 r_{\text{ext}}$$

$$\partial_t r_\times = c_{\text{ext}}\, l\, r_{\text{ext}}\, n - \kappa\, r_\times - c_\times r_\times n + D_\times \nabla^2 r_\times$$

The story: cells consume r_{ext} at rate $c_{\text{ext}} r_{\text{ext}} n$. A fraction l of that flux is leaked as the byproduct $r_\times$; the remaining fraction $(1-l)$ becomes growth. The leaked byproduct is then re-consumed by the same population at rate $c_\times r_\times n$, also contributing to growth.

Step 1: Homogeneous steady state

Set all time derivatives and all Laplacians to zero.

From the r_{ext} equation:

$$K = \kappa r^*_{\text{ext}} + c_{\text{ext}} r^*_{\text{ext}} n^* \quad \Longrightarrow \quad r^*_{\text{ext}} = \frac{K}{\kappa + c_{\text{ext}} n^*}$$

From the $r_\times$ equation:

$$c_{\text{ext}} l\, r^*_{\text{ext}} n^* = \kappa r^*_\times + c_\times r^*_\times n^* \quad \Longrightarrow \quad r^*_\times = \frac{c_{\text{ext}} l\, r^*_{\text{ext}} n^*}{\kappa + c_\times n^*}$$

These two denominators will appear everywhere, so name them now:

$$\boxed{u \equiv \kappa + c_{\text{ext}} n^*, \qquad v \equiv \kappa + c_\times n^*}$$

Each is a total removal rate: u is the rate at which r_{ext} leaves the system (dilution κ plus consumption $c_{\text{ext}} n^*$), and v likewise for $r_\times$. In these terms

$$r^*_{\text{ext}} = \frac{K}{u}, \qquad r^*_\times = \frac{c_{\text{ext}} l K n^*}{uv}$$

From the n equation, the bracket must vanish (taking $n^* \neq 0$):

$$c_{\text{ext}}(1-l)\, r^*_{\text{ext}} + c_\times r^*_\times = m$$

This is the equation that fixes n^*. We won't need to solve it explicitly — but we *will* use the fact that it holds, repeatedly.

Step 2: Perturb

Write

$$n = n^* + \nu\, e^{i\mathbf{k}\cdot\mathbf{x}+\lambda t}, \qquad r_{\text{ext}} = r^*_{\text{ext}} + \rho\, e^{i\mathbf{k}\cdot\mathbf{x}+\lambda t}, \qquad r_\times = r^*_\times + \sigma\, e^{i\mathbf{k}\cdot\mathbf{x}+\lambda t}$$

with ν, ρ, σ infinitesimal. Since $\nabla^2 e^{i\mathbf{k}\cdot\mathbf{x}} = -k^2 e^{i\mathbf{k}\cdot\mathbf{x}}$, every Laplacian becomes multiplication by $-k^2$.

Quasi-steady-state assumption: the resources equilibrate fast, so we set $\partial_t \rho = \partial_t \sigma = 0$ while keeping their diffusion. This is the key simplification: it lets us solve for ρ and σ in terms of ν, and reduces a 3×3 eigenvalue problem to a single scalar equation.

Step 3: Solve for ρ (the external resource response)

Linearise the r_{ext} equation. The only nonlinear term is $-c_{\text{ext}} r_{\text{ext}} n$, which varies as $-c_{\text{ext}}(r^*_{\text{ext}}\nu + n^*\rho)$:

$$0 = -\kappa\rho - c_{\text{ext}}\,(r^*_{\text{ext}}\nu + n^*\rho) - D_{\text{ext}} k^2 \rho$$

Collect the ρ terms:

$$0 = -\underbrace{(\kappa + c_{\text{ext}} n^*)}_{=u}\rho - D_{\text{ext}} k^2 \rho - c_{\text{ext}} r^*_{\text{ext}} \nu$$

$$\boxed{\rho = -\frac{c_{\text{ext}}\, r^*_{\text{ext}}}{u + D_{\text{ext}} k^2}\ \nu}$$

Read it: ρ has the *opposite* sign to ν. More cells means less external resource. This is the inhibitory channel. Its magnitude is suppressed at large k — because rapid diffusion smooths out any local depletion, a short-wavelength bloom cannot dig itself a resource hole.

Step 4: Solve for σ (the byproduct response)

Linearise the $r_\times$ equation. Two nonlinear terms: $+c_{\text{ext}}l\, r_{\text{ext}}n \to c_{\text{ext}}l(r^*_{\text{ext}}\nu + n^*\rho)$, and $-c_\times r_\times n \to -c_\times(r^*_\times \nu + n^*\sigma)$:

$$0 = c_{\text{ext}}l\left(r^*_{\text{ext}}\nu + n^*\rho\right) - \kappa\sigma - c_\times\left(r^*_\times\nu + n^*\sigma\right) - D_\times k^2\sigma$$

Collect the σ terms, which give $-(\kappa + c_\times n^*)\sigma - D_\times k^2\sigma = -(v + D_\times k^2)\sigma$:

$$\sigma = \frac{c_{\text{ext}}l\, r^*_{\text{ext}}\nu \;-\; c_\times r^*_\times \nu \;+\; c_{\text{ext}}l\, n^*\rho}{v + D_\times k^2}$$

Now handle the first two terms. Using $r^*_\times = c_{\text{ext}}lKn^*/(uv)$ and $r^*_{\text{ext}} = K/u$:

$$c_{\text{ext}}l\, r^*_{\text{ext}} - c_\times r^*_\times = \frac{c_{\text{ext}}lK}{u} - \frac{c_\times c_{\text{ext}}lKn^*}{uv} = \frac{c_{\text{ext}}lK}{u}\left(1 - \frac{c_\times n^*}{v}\right) = \frac{c_{\text{ext}}lK}{u}\cdot\frac{\kappa}{v}$$

using $v - c_\times n^* = \kappa$. So

$$\boxed{\sigma = \frac{1}{v + D_\times k^2}\left[\frac{c_{\text{ext}}lK\kappa}{uv}\,\nu \;+\; c_{\text{ext}}l\, n^*\rho\right]}$$

Read it: two contributions.

- The first is **positive**: more cells leak more byproduct. This is the activating channel.
- The second inherits ρ, which is negative. More cells deplete r_{ext}, so *less* gets leaked into $r_\times$. This is an inhibitory contribution travelling through the byproduct.

That second piece is the one that will matter most, so let's track it carefully.

Step 5: The n equation

Linearise. The perturbation ν multiplies the bracket — but **the bracket vanishes at the fixed point** (Step 1). So that term dies, and only n^* times the bracket's perturbation survives:

$$\lambda\nu = n^*\left[c_{\text{ext}}(1-l)\rho + c_\times\sigma\right] - D_n k^2\nu$$

This is the crucial structural fact: n **has no autonomous linear dynamics.** Its entire growth rate is inherited from the resource perturbations. Everything downstream follows from this.

Step 6: Assemble

Substitute σ from Step 4:

$$\lambda = -D_n k^2 + \frac{n^*}{\nu}\left[c_{\text{ext}}(1-l)\rho + \frac{c_\times}{v + D_\times k^2}\left(\frac{c_{\text{ext}} l K \kappa}{uv}\nu + c_{\text{ext}} l\, n^* \rho\right)\right]$$

Now substitute $\rho = -\dfrac{c_{\text{ext}} K/u}{u + D_{\text{ext}} k^2}\nu$ in the two places it appears, and divide out ν:

$$\lambda = -D_n k^2 \underbrace{- \frac{n^* c_{\text{ext}}^2 (1-l) K}{u\,(u + D_{\text{ext}} k^2)}}_{\text{direct competition}} \underbrace{+ \frac{n^* c_\times c_{\text{ext}} l K \kappa}{uv\,(v + D_\times k^2)}}_{\text{cross-feeding gain}} \underbrace{- \frac{n^{*2} c_\times c_{\text{ext}}^2 l K}{u\,(u + D_{\text{ext}} k^2)\,(v + D_\times k^2)}}_{\text{indirect: depletion}\to\text{less leakage}}$$

Step 7: Compact form

Factor each denominator to expose its structure. Define the two **screening lengths**

$$\ell_{\text{ext}} = \sqrt{\frac{D_{\text{ext}}}{u}}, \qquad \ell_\times = \sqrt{\frac{D_\times}{v}}$$

so that $u + D_{\text{ext}} k^2 = u\left(1 + \ell_{\text{ext}}^2 k^2\right)$ and $v + D_\times k^2 = v\left(1 + \ell_\times^2 k^2\right)$. Then, defining

$$A = \frac{n^* c_\times c_{\text{ext}} l K \kappa}{uv^2}, \qquad I_1 = \frac{n^* c_{\text{ext}}^2 (1-l) K}{u^2}, \qquad I_2 = \frac{n^{*2} c_\times c_{\text{ext}}^2 l K}{u^2 v}$$

we arrive at

$$\boxed{\lambda(k) = -D_n k^2 \;+\; \frac{A}{1 + \ell_\times^2 k^2} \;-\; \frac{I_1}{1 + \ell_{\text{ext}}^2 k^2} \;-\; \frac{I_2}{\left(1 + \ell_{\text{ext}}^2 k^2\right)\left(1 + \ell_\times^2 k^2\right)}}$$

All three of A, I_1, I_2 are **positive**. This is an activator–inhibitor dispersion relation: one gain term, two loss terms.

Step 8: Sanity checks

At $k = 0$: all Lorentzians equal 1, so $\lambda(0) = A - I_1 - I_2$. This must be negative for the homogeneous state to be stable — and indeed it equals $n^* G'(n^*)$, the well-mixed eigenvalue. Net feedback is inhibitory in bulk.

At $k \to \infty$ with $D_n = 0$: every term vanishes, $\lambda \to 0$. This is Step 5's consequence: screen out the resources and n has nothing left to do.

Step 9: What drives the instability

Since $\lambda(0) < 0$ and $\lambda(\infty) = 0$ (for $D_n = 0$), instability is decided by *how* zero is approached. Expand at large k:

$$\frac{A}{1+\ell_\times^2 k^2} \sim \frac{A}{\ell_\times^2 k^2}, \qquad \frac{I_1}{1+\ell_{\text{ext}}^2 k^2} \sim \frac{I_1}{\ell_{\text{ext}}^2 k^2}, \qquad \frac{I_2}{(1+\ell_{\text{ext}}^2 k^2)(1+\ell_\times^2 k^2)} \sim \frac{I_2}{\ell_{\text{ext}}^2 \ell_\times^2 k^4}$$

The key observation: I_2 decays as k^{-4}, one power faster than the other two. It is a *double* Lorentzian, because that inhibitory signal must pass through **two** screened resource fields in sequence — depletion of r_{ext}, propagated into $r_\times$. In real space this means its range is $\sqrt{\ell_{\text{ext}}^2 + \ell_\times^2}$, strictly longer than the activator's $\ell_\times$, **for any diffusivities whatsoever**.

So the separation of scales here is **topological, not diffusive** — it comes from path length through the metabolic chain, not from D_{ext} exceeding $D_\times$. Dropping I_2 from the large-k balance:

$$\lambda(k) \approx \frac{1}{k^2}\left(\frac{A}{\ell_\times^2} - \frac{I_1}{\ell_{\text{ext}}^2}\right) = \frac{1}{k^2}\left(\frac{Av}{D_\times} - \frac{I_1 u}{D_{\text{ext}}}\right)$$

Instability requires this to be positive. Substituting the definitions, $Av/D_\times = \dfrac{n^* c_\times c_{\text{ext}} l K \kappa}{uvD_\times}$ and $I_1 u/D_{\text{ext}} = \dfrac{n^* c_{\text{ext}}^2 (1-l) K}{uD_{\text{ext}}}$, so the condition is

$$\boxed{\frac{1}{D_\times} \cdot \frac{c_\times l\, \kappa}{v} \;>\; \frac{1}{D_{\text{ext}}} \cdot c_{\text{ext}}(1-l)}$$

For $D_{\text{ext}} = D_\times$ this reduces to the reaction-only condition $\dfrac{c_\times l \kappa}{\kappa + c_\times n^*} > c_{\text{ext}}(1-l)$ — strong leakage and weak direct competition — with no diffusivity requirement, which is the case you checked against $Kc_{\text{ext}}/(m\kappa) < 1/(1-l)$.

Two necessary ingredients, distinct from each other:

1. **Sign:** the above inequality, which asks whether the net feedback flips positive once the resources are screened. Mostly a statement about reaction kinetics.

2. **Scale:** at least one $D > 0$, so that screening actually happens as k grows. If $D_{\text{ext}} = D_\times = 0$ there is no k-dependence at all and $\lambda(k) \equiv \lambda_0 < 0$ everywhere.

And D_n must be small: it is the only term that *damps* rather than merely screening, and a large enough D_n kills the band regardless of everything else. With $D_n > 0$ the unstable band is finite and a genuine wavelength is selected; with $D_n = 0$ the band extends to $k \to \infty$ and the selected scale must come from elsewhere.